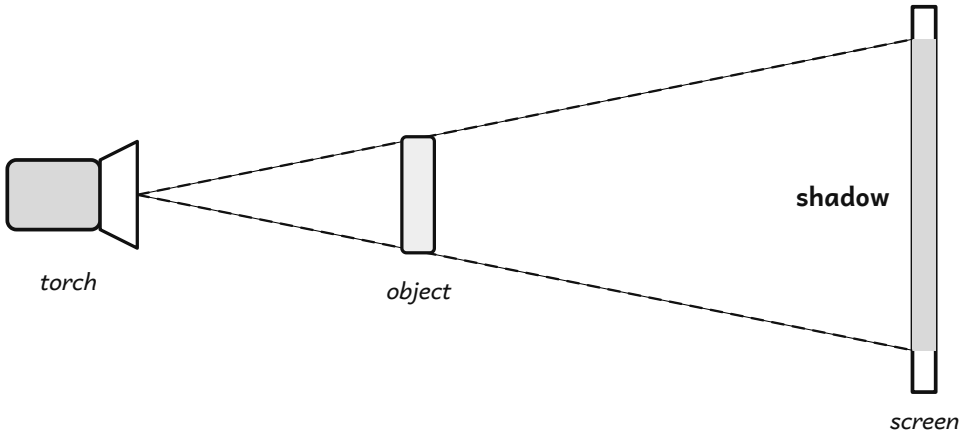
PHYSICAL AND CHEMICAL SCIENCE

Light: Shadows and Intensity

Straight-line travel, transparent, translucent and opaque, shadows, and measuring light



In this pack

Learn it: Shadows and light intensity ★☆☆
Fact check: Light, materials and shadows ★☆☆
Think it through: Changing shadows ★★☆
Investigate: How does distance from the torch change shadow size? ★★☆
Real-world case study: A brighter, cooler classroom ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C8 · SHADOWS

Shadows and light intensity

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Explain that light travels in straight lines and how shadows form.
- Classify materials as transparent, translucent or opaque.
- Explain how shadows change and how light intensity is measured.

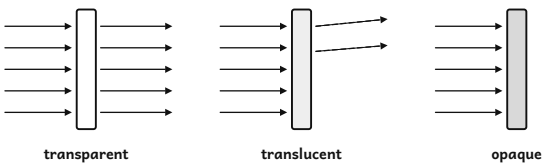
On a sunny afternoon your shadow stretches out long and thin across the school field. At midday it shrinks to a small dark patch around your feet. Why do shadows form, and why do they change?

Light travels in straight lines

Light travels from a source in **straight lines**. You can see this when sunbeams shine through a gap in the curtains into a dusty room, or when a torch beam cuts through mist. It is also why you cannot see round corners.

Transparent, translucent and opaque

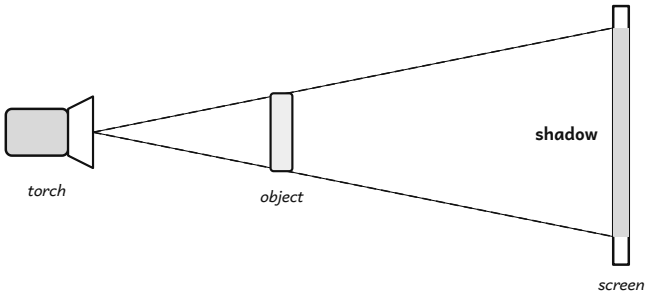
- **Transparent** materials let almost all light pass straight through, so you can see clearly through them: clear glass, clear plastic, water.
- **Translucent** (partly transparent) materials let some light through but scatter it, so things look blurred: tracing paper, frosted glass, thin cotton cloth.
- **Opaque** materials let no light through: wood, cardboard, metal, your body.



How light passes through different materials.

How shadows form

A **shadow** forms when an **opaque** object **blocks** light travelling from a source. Because light travels in straight lines, it cannot bend round the object, so a dark area is left behind it. A shadow has the same **outline shape** as the object, but it shows no colour or detail.



An opaque object blocks light to make a shadow.

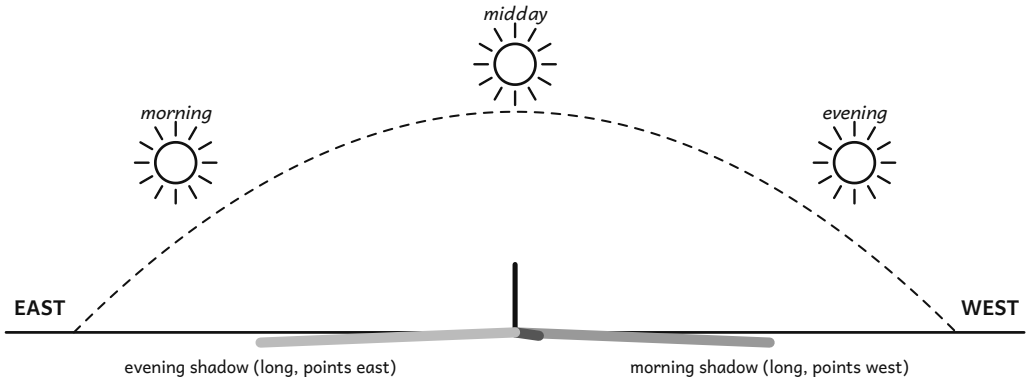
The **closer** an object is to the light source, the **bigger** its shadow, because it blocks more of the light. Moving the object further from the source makes the shadow smaller and sharper. Turning the object changes the shadow's shape.

Bangladesh connection

Shadow puppet shows use cut-out shapes held between a lamp and a cloth screen. In many places, including Bangladesh, the times of some daily prayers were traditionally worked out by measuring the length of shadows.

Shadows during the day

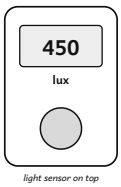
The Sun seems to move across the sky from **east** to **west** (really, the Earth is spinning). In the **morning** and **evening** the Sun is low, so shadows are **long**. At **midday** the Sun is highest, so shadows are **shortest**. Shadows always point ***away*** from the Sun, so they move round during the day. A **sundial** uses this to tell the time.



Shadows of a stick during one day.

Measuring light intensity

Light intensity is how bright light is. It is measured with a **light meter** (or a light sensor on a tablet) in units called **lux**. A light meter gives a number, which is fairer and more accurate than saying "quite bright".



A light meter.

Place	Light intensity (about)
sunny day outdoors	50 000–100 000 lux
cloudy day outdoors	1000–10 000 lux
well-lit classroom	300–500 lux
full moon	less than 1 lux

How scientists measured light

People once thought light arrived instantly. In **1676**, the Danish astronomer **Ole Rømer** noticed that one of Jupiter's moons appeared from behind the planet later than expected when Earth was further away. He realised that light **takes time** to travel. In **1849**, **Hippolyte Fizeau** in France measured the speed of light on Earth using a spinning toothed wheel and a mirror 8 km away. Today we know light travels about **300 000 km every second**. Sunlight takes about 8 minutes to reach us.

Key words

transparent	lets light through clearly
translucent	lets some light through but scatters it
opaque	lets no light through
shadow	a dark area where light is blocked
light intensity	how bright light is
lux	the unit of light intensity
light meter	an instrument that measures light intensity
sundial	a clock that uses a shadow

FACT CHECK · C8 · SHADOWS

Light, materials and shadows

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Classify materials.
- Recall how shadows form and change.

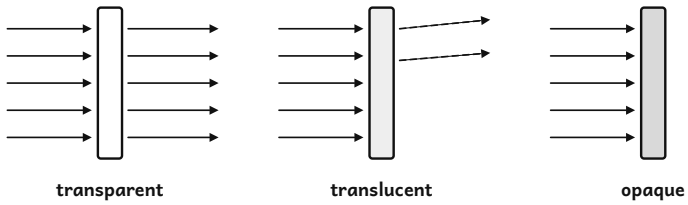
1 **Sort the materials.** [9]

SORT THESE

clear plastic bottle **tracing paper** **water** **clear glass** **steel tray** **cardboard** **wood** **thin cotton cloth** **frosted glass**

Transparent	Translucent	Opaque

2 **Why can you read a book through clear glass but not through tracing paper or cardboard?** [3]



3 **Fill in the gaps.** [5]

WORD BANK

longest **lux** **curved** **opaque** **metres** **shortest** **straight**

Light travels in _____ lines. A shadow forms when an _____ object blocks the light. Shadows are _____ in the morning and evening and _____ at midday. Light intensity is measured in _____.

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. A shadow shows the colours of the object.			
2. Moving an object closer to the torch makes its shadow bigger.			
3. Shadows point towards the Sun.			
4. A light meter measures how bright light is.			

THINK IT THROUGH · C8 · SHADOWS

Changing shadows

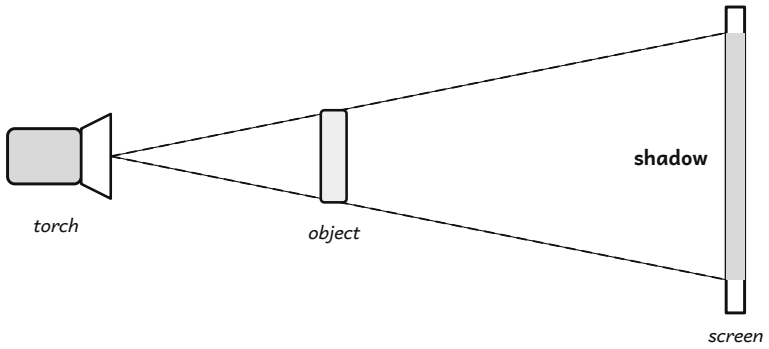
★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain shadow size and shape.
- Interpret shadow data from a day.

1 **Look at the torch, object and screen.** [4]



a) What would happen to the shadow if you moved the object closer to the torch?

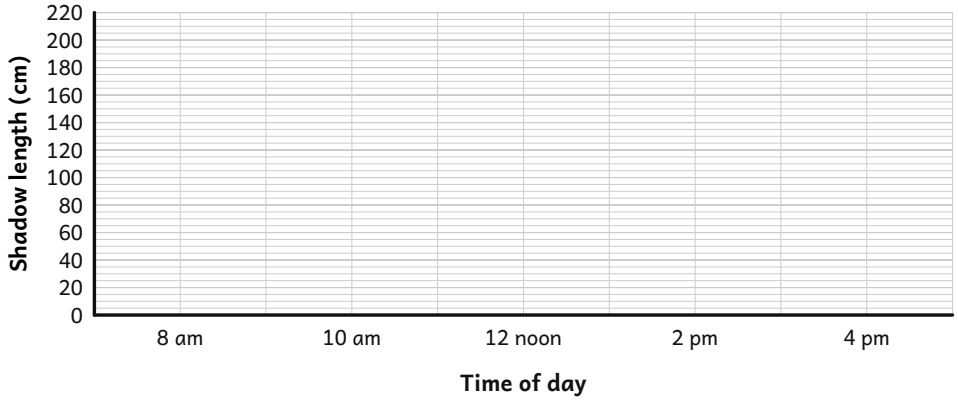
b) How could you make the shadow a different shape without changing the object?

c) Why is the shadow the same shape as the object?

2 **Why can you not see a clear shadow on a very cloudy day?** [1]

3 **Nadia measured the shadow of a 1 metre stick in the school yard.** [6]

Time	8 am	10 am	12 noon	2 pm	4 pm
Shadow length (cm)	210	80	30	75	190



a) Draw a bar chart of the results.

b) Describe the pattern and explain it.

.....

.....

.....

c) In which direction did the shadow point at 8 am? Why?

.....

.....

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How does distance from the torch change shadow size?

★★☆ Grade 5

Name: **Class:** **Date:**

Learning goals

- Plan a fair test.
- Plot a line graph.
- Use the pattern to predict.

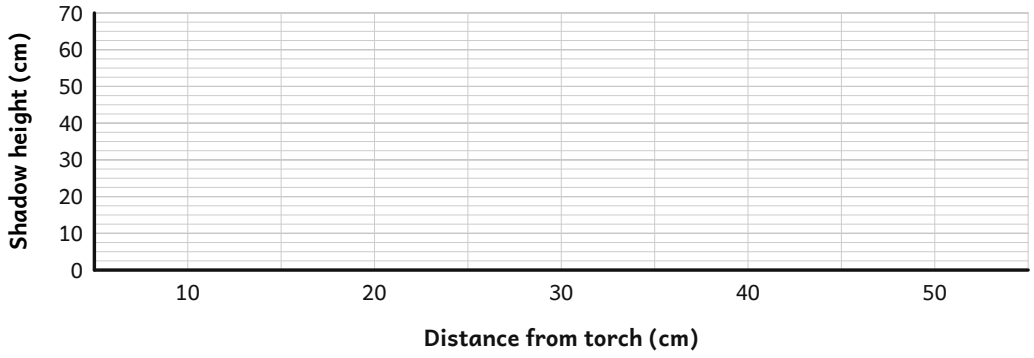
Jamal put a torch on a table and a screen 60 cm away. He held a card puppet 10 cm tall at different distances from the torch and measured the height of the shadow on the screen.

1 **Complete the plan.** [5]

Question	
Change	
Measure	
Keep the same	
Prediction	

2 **Plot the results as a line graph.** [4]

Distance from torch (cm)	10	20	30	40	50
Shadow height (cm)	60	30	20	15	12



3 **Describe the pattern. Did the results support Jamal's prediction?** [2]

.....

.....

4 **Predict the shadow height when the puppet is 15 cm from the torch.** [1]

.....

5 **Why can the shadow never be smaller than 10 cm in this test?** [1]

.....

REAL-WORLD CASE STUDY · C8 · SHADOWS

A brighter, cooler classroom

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Use light intensity data.
- Apply knowledge of materials to a design problem.
- Weigh up costs and benefits.

Science · Design · Maths · Environment · Economics

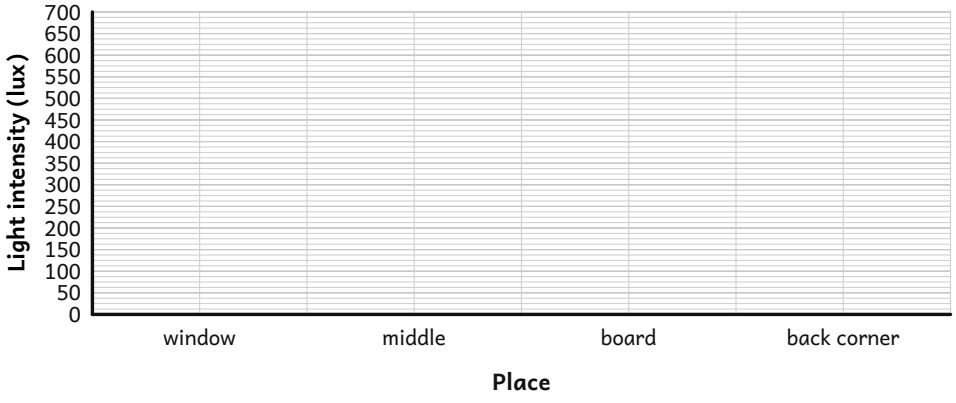
In Bangladesh, classrooms can get very **hot** from April to October. Teachers often close the curtains to keep out the heat and glare, but then the room gets **dark** and the lights must be switched on, which uses electricity. Some learners at the back cannot see the board clearly.

Experts recommend about **300 to 500 lux** for reading and writing in classrooms. Too little light strains the eyes; direct sunlight on desks causes **glare** and heat. Good classroom design uses daylight wisely: light-coloured walls and ceilings **reflect** light around the room, **translucent** blinds let light in but block direct sun, and trees or shades outside block heat from the walls and windows.

Example readings below; lighting levels based on common school lighting guidelines.

1 **Grade 5 measured light intensity around their classroom with a light meter app at 11 am, with the curtains closed and lights off.** [3]

Place	by the window	middle desk	board	back corner
Light (lux)	650	280	220	120



a) Draw the bar chart.

b) Which places are below 300 lux?

2 **For each idea, explain the science.** [3]

Idea	How it helps (use the words reflect, translucent or opaque)
paint the walls white	
translucent blinds	
plant trees outside the windows	

3 **What might each person think?** [3]

Person	View
the head teacher	
a learner at the back	
the school caretaker	

4 **Draw and label your design for a brighter, cooler classroom.** [3]

My classroom design

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

I	S	O	S	U	N	D	I	A	L	F
T	C	T	U	O	U	Y	F	P	B	C
W	S	K	R	K	I	M	P	T	F	V
N	G	E	G	A	W	V	O	O	N	W
O	U	M	E	S	I	C	R	R	R	O
O	Y	P	A	H	O	G	H	C	D	P
N	I	M	G	A	B	B	H	H	V	A
P	J	E	E	D	L	L	I	T	D	Q
T	A	T	C	O	O	W	K	E	K	U
N	E	E	Y	W	C	R	K	N	R	E
T	P	R	O	M	K	I	T	L	U	X

block lux meter noon opaque shadow straight sundial torch

2 **Which material would make the darkest shadow?** [1]

- ◯ clear glass
- ◯ tracing paper
- ◯ cardboard
- ◯ water

3 **When is your shadow shortest on a sunny day?** [1]

- ◯ early morning
- ◯ midday
- ◯ late afternoon
- ◯ at sunset

4 **Draw the shadow of the stick at 9 am and at 4 pm. Label east and west.** [2]

Stick shadows

5 **Challenge. How did Ole Rømer show that light takes time to travel? Why was this an important idea?** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that light travels in straight lines	○	○	○
sort materials as transparent, translucent or opaque	○	○	○
explain how a shadow forms	○	○	○
investigate how shadow size changes	○	○	○
explain how shadows change during the day	○	○	○
measure light intensity in lux	○	○	○